DINOJAN VIVEKANANTHARASA

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PROFESSIONAL SUMMARY

A Computer Engineering graduate with hands-on experience in **embedded systems, sensor integration, and low-level programming**. Experienced in building sensor-driven microcontroller control systems with automatic safety cut-offs, analog circuit design, and performance-critical C++ development. Combines hardware fundamentals with one year of professional software engineering and HPC, simulation, and ML projects.

EDUCATION

BSc.Eng (Honours) in Computer Engineering **Feb 2022 – Sep 2026**
Faculty of Engineering, University of Ruhuna, Galle, Sri Lanka

- **Certificate Course in Soft Skills Development** (2022–2024): communication, leadership, problem solving

GCE Advanced Level **Oct 2020**
V/Vavuniya Tamil Madhya Maha Vidyalayam *Physical Science: A, C, S*

PROFESSIONAL EXPERIENCE

Yarl Ventures (PVT) Ltd, Jaffna **May 2025 – May 2026**
Full Stack Developer *Dec 2025 – May 2026*

- Built viewer-admin access and clinic management in MediMan; managed its releases on Linux servers.
- Contributed to EDUS LMS payments, wallets, chat, role-permission testing, and API optimisation.

Full Stack Developer Intern *May 2025 – Nov 2025*

- Built MediMan admin features with server-side pagination and field projection to cut API payload sizes.
- Integrated Twilio SMS/OTP, Nodemailer, and Firebase push; debugged production via CloudWatch logs.

TECHNICAL SKILLS

Embedded & Hardware: Arduino, C/C++, UART, I2C, Sensor Interfacing
Electronics: Analog Circuit Design, BJT Amplifiers, Proteus Simulation
Programming: C++17, Python, Bash
Systems & Tools: Linux, MPI, OpenMP, CUDA, NS-3, Git

PROJECTS

Smart Home Automation with Emergency Power Shutdown **Jun 2024 – Nov 2024**
Arduino (ATmega328P), C++, MQ-2, LDR, Relays, I2C LCD, Proteus

- Built the sensors, control logic, GSM simulation, and hardware of an Arduino smart-home system.
- A detected gas leak cuts the power through a relay automatically, alongside PWM fan and LDR lighting.

Parallel Graph Processing (HPC) **Dec 2025 – Apr 2026**
C++17, MPI, OpenMP, CUDA

- Built parallel BFS in OpenMP, MPI, and CUDA, plus the benchmarks and the performance report.
- All versions match the serial baseline exactly; Amdahl analysis gives an 89.4% parallel fraction.

TRACES – Secure Vehicular Network Simulation **Dec 2025 – Sep 2026**
NS-3, Python, Post-Quantum Cryptography

- Final-year research: built the 200-vehicle NS-3 network, LLM detection, and post-quantum layer.
- Vehicle links secured with ML-KEM-1024 and Falcon-1024, with multi-stage LLM anomaly detection.

Common-Emitter BJT Amplifier **May 2025**

Analog Electronics, Circuit Design, Proteus, PCB Layout

- Designed a common-emitter amplifier: calculations, Proteus simulation, PCB layout, and build.
- Hit the target gain of 185 after tuning RE and C3, with 5.56 MHz bandwidth against a 5.5 MHz target.

Real-Time Fleet Operations Platform **May 2026 – Aug 2026**

Python, Kafka, Spark Structured Streaming, Airflow, Prometheus, Grafana, Docker

- Streams telemetry from 12 simulated vehicles (about 6 events/s) through Kafka into Spark and a data lake.
- An Airflow batch layer reconciles daily costs exactly; Prometheus alerts and Grafana track pipeline health.

IoT Room Occupancy Detection **Jun 2024 – Nov 2024**

Python, Scikit-learn, Sensor Data

- Occupancy detection from environmental IoT sensor readings with Logistic Regression and Decision Trees.
- Decision Tree reached 99.1% and Logistic Regression 98.8% test accuracy (F1 0.991 and 0.988).

Healthcare Mobile Application **Jun 2024 – Nov 2024**

Flutter, Dart, Firebase

- Built the doctor and patient screens, authentication, and Firebase sync of a Flutter healthcare app.
- Doctors manage availability, appointments, prescriptions, and patient history in one mobile app.

PUBLICATIONS

G. S. J. Venujan, M. F. A. Munsif, **V. Dinojan**, T. Jegan, and G. G. N. Sandamali, "AutoGradePro: An Automated Grading System for Short-Answer Assessments Using Rule-Based Methods and Large Language Models," in *Proceedings of the Vice Chancellor's Awards and Academic Sessions 2026*, University of Ruhuna, Sri Lanka, Mar. 2026, Abstract ABS-064, p. 35. ISSN 2362-0412.

VOLUNTEER EXPERIENCE

Member **Apr 2024 – Present**

IESL, Galle

Member **Sep 2023 – Dec 2024**

IEEE Student Branch, Ruhuna, Galle

Treasurer **Apr 2023 – Apr 2024**

All University Students' Development Association, Vavuniya

Senior Prefect **Jul 2017 – Jul 2018**

V/V.T.M.M.V, Vavuniya

ACHIEVEMENTS

- Participated in **IEEE Xtreme 18.0 (2024)**, a global 24-hour algorithmic coding contest (3-member team)
- Competed in **IEEE INSL Southern Provincial Competition**, 30th October 2024
- **3rd Runner-Up**, Provincial Level Under-19 Boys Chess, 2015 and 2016
- Participated in the **Galle Division FIDE 100 Commemorative Chess Event** (Open Category), 2024

REFEREES

- **Dr. Kushan Sudheera**
  Senior Lecturer
  Department of Electrical and Information Engineering, Faculty of Engineering, University of Ruhuna
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- **Dr. Nadeesha Sandamali**
  Senior Lecturer
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